RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Date: 02/05/2025	**CIE - 1**	**Max. Marks:100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

Department of Mathematics

	Quiz			
SL.No	**Questions**	**M**	**BT**	**CO**
1	In a certain country, the growth in the number of computer scientists is observed to follow the law $\frac{dx}{dt} = \frac{1}{15x}$. If initially there were 1000 of them, then after 15 years, their approximate number will be _____ and their doubling period is approximately _____ years.	**2**	**L1**	**2**
2	A body of mass 3 kg is projected at an angle of 30° from the ground with an initial velocity of 25 m/s, acceleration due to gravity is $g \simeq 10\, m/s^2$. The maximum horizontal distance is_____.	**2**	**L1**	**1**
3	A cup of coffee at 90°C is left in a room at 25°C. If the cooling constant is $k = 0.05$, using Newton's law of cooling, the temperature of the coffee after 10 minutes is ____.	**2**	**L2**	**3**
4	Find the maximum height of the projectile at the time $v\ \frac{sin\alpha}{2g}$.	**2**	**L2**	**2**
5	Discuss about Fick's law of diffusion model.	**2**	**L1**	**1**
SL. No	**Test**	**M**	**BT**	**CO**
1.	Develop a differential equation model based on the logistic law of population growth, solve it and analyse any one real life scenario.	10	L3	2
2.	Describe the systematic 12-step procedure for tackling a problem through the use of mathematical modelling.	10	L1	1
3.	Model and describe the second-order differential equations representing a mass-spring-dashpot mechanical system. Also highlight the analogies between the elements of the mechanical and electrical systems.	10	L2	2
4.	A nutrient is periodically injected into a patient's bloodstream at fixed intervals of time T. Meanwhile, the nutrient is continuously absorbed by the body at a rate proportional to its concentration $c(t)$ in the blood at time t. Derive the differential equation governing $c(t)$. Solve it and interpret the long-term behaviour of the nutrient concentration in the bloodstream.	10	L4	4

5.	i)	Model the motion of a projectile launched in a vacuum with an initial velocity at a given horizontal angle. Formulate the governing equations of motion, solve them and determine key characteristics such as time of flight, maximum height and horizontal range.	10	L3	3
	ii)	Explain any two characteristics of Mathematical modelling.			

Date: 02/05/2025	**Test - 1**	**Max. Marks:100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

SCHEME & SOLUTIONS

S No	Solutions with Scheme	Marks
	Part-A	
1.	$x(15) = 1000 \cdot e^{\frac{15}{15}} = 1000 \cdot e = 1000 \cdot 2.718 = 2718$	1+1
2.	$x = \frac{V^2 \sin 2\alpha}{g} = 25\frac{\sin 60^0}{10} = 54.13m$.	1+1
3.	$T(10) = 25 + (90 - 25)e^{-0.05 \cdot 10} = 25 + 65 \cdot e^{-0.5} = 0.6065$	1+1
4.	$y = tV\sin(\alpha) - \frac{gt^2}{2} = \frac{3}{8g}V^2 \sin^2 \alpha$	1+1
5.	According to Fick's law of diffusion, the rate of movement of a solute across a thin membrane is proportional to the area of the membrane and to the difference in concentrations of the solute on one slide is kept fixed at a and the concentration of the solution on the other side initially $c_0 > a$ then $\frac{dc}{dt} = k(a - c), a - c = \big(a - c(0)\big)e^{-kt}$ $\Rightarrow c(t) \to a$ as $t \to \infty$ whatever be the value of $c(0)$.	2

Q.No	Part-B	Marks
1	As the population increases, due to overcrowding and limitations of resources, the birth rate b decreases and the death rate d increases with the population size x. $B = b_1 - b_2x,\ D = d_1 + d_2x,\ \ b_1, b_2, d_1, d_2 > 0 \quad \frac{dx}{dt} = (B - D)x = [b_1 - b_2x - (d_1 + d_2x)]x$ $\Rightarrow \frac{dx}{dt} = (a - bx)x - - - - - - - (1), \frac{dx}{(a-bx)x} = dt$ By integrating , $\ln\left(\frac{x}{(a-bx)}\right) = at + c \Rightarrow \frac{x}{a-bx} = Ce^{at} \Rightarrow \frac{x(0)}{a-bx(0)} = C$ $\therefore \frac{x(t)}{a - bx(t)} = \frac{x(0)}{a - bx(0)}e^{at} - - - - - (2)$ From (1) and (2) , we have	2+2+ 2+2+2

$$a - bx(0) > 0 \Rightarrow \frac{a}{b} > x(0) \Rightarrow a - bx(t) > 0 \Rightarrow \frac{a}{b} > x(t) \Rightarrow \frac{dx}{dt} > 0$$

$\Rightarrow x(t)$ is a monotonic increasing function of t which approaches $\frac{a}{b}$ as $t \to \infty$

$a - bx(0) < 0 \Rightarrow \frac{a}{b} < x(0) \Rightarrow a - bx(t) < 0 \Rightarrow \frac{a}{b} < x(t) \Rightarrow \frac{dx}{dt} < 0$

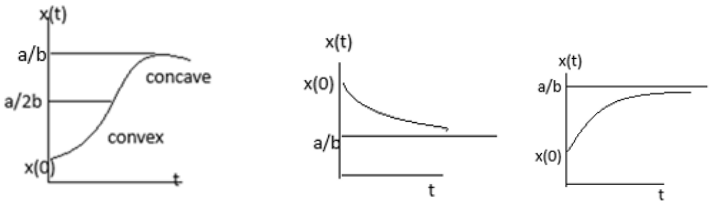
$\Rightarrow x(t)$ is a monotonic decreasing function which approaches $\frac{a}{b}$ as $t \to \infty$

$\Rightarrow \frac{d^2x}{dt^2} = a - 2bx \Rightarrow \frac{d^2x}{dt^2} \lesseqgtr 0$ according as $x \gtreqless \frac{a}{2b}$

The growth curve is convex if $x < \frac{a}{2b}$, concave if $x > \frac{a}{2b}$ and has a point of inflexion at $x = \frac{a}{2b}$.

If $x(0) < \frac{a}{2b}, x(t)$ increases at an increasing rate till $\frac{a}{2b}$, then it increases at a decreasing rate and approaches $\frac{a}{b}$ as $t \to \infty$.

If $\frac{a}{2b} < x(0) < \frac{a}{b}$, then $x(t)$ increases at a decreasing rate and approaches $\frac{a}{b}$ as $t \to \infty$.



Spread of technological innovations and infectious diseases:

Let $N(t)$ be the number of companies which have adopted a technological innovation till time t, then the rate of change of

N depends on the number of companies which have adopted this innovation and the number of those which have not yet adopted it. If R is the total number of companies in the region, then $\frac{dN}{dt} = kN(R - N) - - - - -$ (3)

Which is the logistic law. This shows that ultimately, all companies would adopt this innovation.

If R is the total number of persons in a system , and if $N(t)$ is the number of infected persons, then equation (3) represents spreading of an infectious disease.

Note: In the above examples(situations) , N(t) is essentially an integer valued variable , but we have treated it as a continuous one . this can be regarded as a n idealisation of the situation or as an approximation to reality.

Law of mass action: Chemical reactions:

Two chemical substances combine in the ratio a:b to form a third substance z

If z(t) is the amount of the third substance at time t, then a proportion $\frac{a}{a+b}z(t)$ of it consists of the first substance and a proportion of $\frac{b}{a+b}z$(t)

of second.

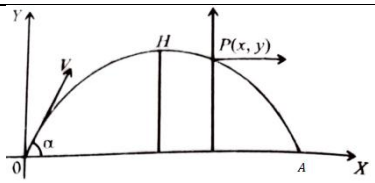
The rate of formation of third substance is proportional to the product of the amount of the two component substances which have not yet combined together.

If A and B are the initial amounts of the two substances, then we get

$$\boldsymbol{\frac{dz}{dt} = k(A - \frac{a}{a+b}z)(B - \frac{b}{a+b}z)}$$

	This is a nonlinear DE for a second order reaction. For the nth order reaction we get the non-linear equation as $\boldsymbol{\frac{dz}{dt} = k(A_1 - a_1 z)(A_2 - a_2 z) \ldots \ldots (A_n - a_n z)\,, a_1 + a_2 + \cdots + a_n = 1.}$	
2	twelve-point procedure for solving problems through mathematical modeling. Be clear about the real-world situation to be investigated. Find all its essential characteristics relevant to the situation and the aspects that can be ignored. (ie of minimal relevance) Think about all the physical, chemical, biological, social, economic laws that may be relevant to the situation. Formulate the problem into problem language (PL) Think about all the variable $x_1, x_2, \ldots x_n$ and the parameters $a_1, a_2, \ldots a_m$ involved. Classify them into known and unknown ones. Choose the most appropriate model and translate the problem suitably into Mathematical Language (ML) Think of all possible ways of solving the problem. The methods may be analytical (Preferably), numerical or simulation. If a reasonable change in the assumptions makes analytical solution possible, investigate the possibility. Make an error analysis of the method used. If the error is not within acceptable limits, change the method of solution. Translate the final solution into PL Compare the predictions with available observations or data. If the agreement is good, accept the model else change the assumptions and approximations in the light of the discrepancies observed and proceed as before. Continue the process till a satisfactory model is obtained. Deduce the conclusions from the model and test these conclusions against easier data and additional data that may be collected and see if the agreement still continues to be good.	**1+2 +2+2 +1+2**
3	Consider an ordinary coil spring suspended from a fixed support. Attach a body at its lower end, for instance, an iron piece at rest. We choose the downward direction as positive. Let the metal piece be pull down by an amount $y > 0$, then by hook's law, the equation of motion is $my'' = -ky,\ (k > 0)$ where $k\ (> 0)$ is the spring constant. Thus $my'' + ky = 0$ $y(t) = A\cos w_0 t + B\sin w_0 t$ where $w_0 = \sqrt{\frac{k}{m}}$ Now add a damping force proportional to the velocity obtaining $my'' + cy' + ky = 0$ This can be done by connecting the ball to a dashpot where c (>0) is the damping constant. Here $\lambda_1 + \lambda_2 = -\frac{c}{m}$ and $\lambda_1\lambda_2 = \frac{k}{m}$ Case(i): over damping If the damping constant c is so large that $c^2 > 4mk$, then the roots λ_1 and λ_2 are real, distinct and negative. $y(t) = A_1\, e^{\lambda_1 t} + A_2 e^{\lambda_2 t}$ Thus $y(t) \to 0$ as $t \to \infty$ and the motion is said to be overdamped. Case (ii): Critical damping If $c^2 = 4mk$, then roots $\lambda_1 = \lambda_2 = -\frac{c}{2m}$, $y(t) = (A_1 + A_2 t)e^{-\frac{c}{2m}t}$ Again $y(t) \to 0$ as $t \to \infty$ and eventually the body comes to rest, but not as quickly as case (i) because of the first term which is a polynomial. The motion is said to be critically damped. Case (iii): Under damping If $c^2 < 4mk$, then the roots are imaginary say $\alpha \pm i\beta$. Then $y(t) = e^{\alpha t}(A_1\cos\beta t + A_2\sin\beta t)$ This means keeps oscillating but since $(\alpha = -\frac{c}{2m})$, the oscillations are damped out and tend to zero as $t \to \infty$. The motion is said to be underdamped.	**2+2 +2+2 +2**

	This model is extended by including an external force $r(t)$ acting on the body. Then we have $my'' + cy' + ky = r(t)$ ----------(5) periodic external forces of the form $F\cos\omega t$, $my'' + cy' + ky = F\cos\omega t$. Both the equations (4)(in LRC circuit) and (5) (from mass-spring-dashpot) can be compared which gives the following correspondences. Mass $m \leftrightarrow$ inductance L, Frition coefficient $C \leftrightarrow$ resistance R,Spring constant $K \leftrightarrow$ inverse capacitanc $\frac{1}{C}$, Impressed force $r(t) \leftrightarrow$ impressed voltage $E(t)$,Displacement $y \leftrightarrow$ charge q Velocity $v \leftrightarrow$ current i	
4	let a dose D of a medicine be given to a patient at regular intervals of duration T each. The medicine also disappears from the system at a rate proportional to $c(t)$, the concentration of the medicine in the blood stream Then by continuity principle $V\frac{dc}{dt} = -kc$ Integrating we get $c(t) = De^{-\frac{k}{V}t} \quad 0 \le t < T$ $-------- (4)$ At time $T(t \ge T)$, the residue of the first dose is $De^{-\frac{k}{V}t}$ and now another dose D is given so that we get $c(t) = De^{-\frac{k}{V}t} + De^{-\frac{k}{V}(t-T)} \quad T \le t < 2T$ The first term gives the residual of the first dose and the second term gives the residual of the second dose. Proceeding in this way after n doses we get $c(t) = De^{-\frac{k}{V}t} + De^{-\frac{k}{V}(t-T)} + De^{-\frac{k}{V}(t-2T)} \pm ----- De^{-\frac{k}{V}(t-(n-1)T)}$ $= De^{-\frac{k}{V}t}\left[1 + e^{\frac{k}{V}T} + e^{\frac{2k}{V}T} + ----- e^{(n-1)\frac{k}{V}T}\right] = De^{-\frac{k}{V}t}\frac{\left(e^{\frac{nk}{V}T}-1\right)}{e^{\frac{k}{V}T}-1}, (n-1)T \le t < nT$ Just before the next dose $c(nT-0) = De^{-\frac{nk}{V}T}\left[\frac{\left(e^{n\frac{k}{V}T}-1\right)}{e^{\frac{k}{V}T}-1}\right] = \frac{D\left(1-e^{-n\frac{k}{V}T}\right)}{e^{\frac{k}{V}T}-1}$ When next dose is given $c(nT+0) = De^{-\frac{nk}{V}T}\frac{\left(e^{n\frac{k}{V}T}-1\right)}{e^{\frac{k}{V}T}-1} + D = \frac{D\left(e^{\frac{k}{V}T}-e^{-n\frac{k}{V}T}\right)}{e^{\frac{k}{V}T}-1}$ $As\ n \to \infty$ we have $D\frac{e^{\frac{k}{V}T}-0}{e^{\frac{k}{V}T}-1}$, $\frac{D}{\frac{\left(e^{\frac{k}{V}T}-1\right)}{e^{\frac{k}{V}T}}} = \frac{D}{1-e^{-\frac{k}{V}T}}$. The concentration never exceeds $\frac{D}{1-e^{-\frac{k}{V}T}}$. In each interval concentration decreases. In any interval, the concentration is maximum at the beginning and that maximum concentration keep increasing interval by interval but will be below $\frac{D}{1-e^{-\frac{k}{V}T}}$ The minimum value in an interval occurs at the end of each interval. This also keep increasing but lies below $\frac{D}{e^{\frac{k}{V}T}-1}$. The concentration curve is piecewise continuous and has points of discontinuity at $T, 2T, 3T, \ldots..$ By injecting glucose on penicillin in blood and fitting curve (4) to the data, we can estimate the value of k and V. In particular this gives a method for finding the volume of	**1+2 +2+2 +1+2**

	blood in the human body.	
5	 A particle of mass m is projected from the origin in vacuum with velocity V inclined at angle α to the horizontal. Suppose at time t, it is at position P $(x(t), y(t))$ and horizontal and vertical velocity components are $u(t), v(t)$ respectively, then the equations of motion are $m\frac{du}{dt} = 0$ and $m\frac{dv}{dt} = -mg$ Integrating, we get $u(t) = c_1$ and $v(t) = -gt + c_2$, $u(0) = Vcos(\alpha) = c_1$ and $v(0) = V\sin(\alpha) = -g(0) + c_2$ $\Rightarrow c_1 = Vcos(\alpha)$ and $c_2 = V\sin(\alpha)$, $\therefore u = Vcos(\alpha)$ and $v = V\sin(\alpha) - gt$ $\frac{dx}{dt} = Vcos(\alpha)$ and $\frac{dy}{dt} = V\sin(\alpha) - gt$, $x = Vcos(\alpha)t$ and $y = tV\sin(\alpha) - \frac{gt^2}{2}$ ----------(1) $y = xtan(\alpha) - \frac{1}{2}\frac{gx^2}{v^2\cos^2(\alpha)}$, $y = x\left(tan(\alpha) - \frac{1}{2}\frac{gx}{v^2\cos^2(\alpha)}\right)$. The parabola cuts $y = 0$ when $x = 0$ or $gx = 2V^2\cos^2\alpha\tan\alpha$, i.e, $x = \frac{V^2\sin 2\alpha}{g}$ The range of the particle is $R = \frac{V^2\sin 2\alpha}{g}$ Putting $y = 0$ in (1), $t = 0$ or $t = \frac{2V\sin\alpha}{g}$ this gives the time of flight T. **Characteristics of mathematical models:** Realism of model: Trade-off between “mathematically tractable” and “realistic”. Robustness: A MM is said to be robust if small changes in parameter lead to only small changes in the behaviour of the model. Generality and Applicability of models: Some Models are applicable to a wide variety of situations while others are applicable to specific situation only. Dictionary of mathematical models: It is unlikely that we shall ever have a complete dictionary of MM. Familiarity with existing models will always be useful, but new situations will always demand construction of new models. Unity of disciplines: When a Number of different situations are represented by the same mathematical model, it reveals a certain identify of structures of these situation. It can lead to a certain economy of efforts and it can reveal a certain underlying unity between different disciplines.	**+2** **+2+2** **+2**

SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Date: 02/05/2025	**Test - 1**	**Max. Marks:100**
Semester: VI	**UG**	**Duration: 2 Hrs. (2pm-4pm)**
Course Title: Mathematical Modelling		**Course Code: MA266TEU**

Declaration by the Course handling faculties:

As a teaching faculty of the course: **Mathematical Modelling** with code: **MA266TEU**

We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analysing, evaluating, and creating.	
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	
5	Course handling faculty (as applicable) who are responsible for preparing the question paper, Scheme and Solutions have unanimously agreed to utilize this Question paper for conducting the Continuous Internal Evaluation.	
6	The Question paper, Scheme and complete Solutions have been submitted to the Test coordinators within the designated time-frame to ensure the smooth conduction of Continuous Internal Evaluations	

Course handling Faculties:

Name: **Signature**

1. Dr. Y. Sailaja

2. Dr. Nivya Muchikel

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	**Awarded**
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme & complete Solutions in the format	10	
	Total points	**50**	
Any other comments by the scrutinizer :			

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	**Signature of Course coordinator**
Accepted/Rejected	**Signature of Scrutinizer** **(Name:)**

Signature of HOD